

EXAM 7 – SPRING 2015

6. (2.75 points)

Given the following information for a book of workers compensation business:

Reported Losses (\$000) – \$250,000 Per-Occurrence Limit

<u>Accident Year</u>	<u>12 Months</u>	<u>24 Months</u>	<u>36 Months</u>	<u>48 Months</u>
2011	6,000	8,472	11,642	12,860
2012	5,620	8,748	12,156	
2013	6,482	9,598		
2014	6,216			

Reported Losses (\$000) – \$1,000,000 Per-Occurrence Limit

<u>Accident Year</u>	<u>12 Months</u>	<u>24 Months</u>	<u>36 Months</u>	<u>48 Months</u>
2011	6,798	12,041	19,888	24,106
2012	5,823	10,541	17,896	
2013	7,321	13,877		
2014	6,984			

2014 earned premium	\$44,622,000
Expected loss ratio assumed in pricing	60%
Industry per-occurrence charge at \$250,000	0.47
Industry per-occurrence charge at \$1,000,000	0.05
Increased limit factor from basic limit to \$250,000	1.21
Increased limit factor from basic limit to \$1,000,000	2.27

- Losses are fully developed at 48 months.
- Ratio of reported losses limited to \$250,000 per occurrence to reported losses limited to \$1,000,000 per occurrence is given by $y = (\text{ultimate ratio}) e^{0.177t}$ where t is the time to ultimate maturity, measured in years.

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(6 continued)

a. (0.5 point)

Using Siewert's loss ratio approach, estimate ultimate losses in the layer between \$250,000 and \$1,000,000 for accident year 2014.

b. (1.25 points)

Using Siewert's implied development approach, estimate ultimate losses in the layer between \$250,000 and \$1,000,000 for accident year 2014.

c. (1 point)

By adjusting the loss development factors for losses limited to \$1,000,000 per occurrence to apply to losses in the layer between \$250,000 and \$1,000,000, calculate ultimate losses in the layer between \$250,000 and \$1,000,000 for accident year 2014.

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QUESTION 6

TOTAL POINT VALUE: 2.75

LEARNING OBJECTIVE: A4: Estimate unpaid claims for various layers of claims.

SAMPLE ANSWERS

Part a: 0.5 point

Sample Answer 1

Ult limited @ 250,000 = 44,622M x 0.6 x (1 – 0.47) = 14, 189,796

Ult limited @ 1M = 44,622M x 0.6 x (1 – 0.05) = 25,434,540

Ult. layer 250000 to 1,000,000 = 25,434,540 – 14,189,796 = 11,244,744

Sample Answer 2

LR approach

ultimate at 1,000,000 limit = (44,622,000)(.6)(.95)=25,434,540

Ultimate at 250,000 limit = (44,622,000)(.6)(.53)=14,189,796

Difference = 25,434,540 – 14,189,796 = \$11,244,744

Part b: 1.25 points

Sample Answer 1

	12-24	24-36	36-48
LDF at 250k	1.481	1.382	1.105
To Ultimate	2.262		

Ultimate at 250k = 6,216 x 2.262 = 14,061k

	12-24	24-36	36-48
LDF at 1M	1.828	1.673	1.212
To Ultimate	3.707		

Ultimate at 1M = 6,984 x 3.707 = 25,890k

Ultimate in layer = (25,890 – 14,061) x 1,000 = 11,829,000

Sample Answer 2

At 250k limit:

LDF(1) = (8,472+ 8,748 + 9,598) / (6,000 + 5,620 + 6,482) = 1.4815

LDF(2) = (11,642 + 12,156) / (8,472 + 8,748) = 1.382

LDF(3) = 12,860 / 11,642 = 1.1046

Ultimate AY loss @ 250k = 6,216 x 1.4815 x 1.382 x 1.1046 = 14,058

At 1M limit:

LDF(1) = (12,041 + 10,541 + 13,877) / (6,798 + 5,823 + 7,321) = 1.828

LDF(2) = (19,888 + 17,896) / (12,041 + 10,541) = 1.673

LDF(3) = 24,106 / 19,888 = 1.212

Ultimate AY loss @1M = 6,984 x 1.828 x 1.673 x 1.212 = 25,887

Implied Ultimate AY 2014 loss between 250-1M = 25,887 – 14,058 = 11,829 (000s)

Part c: 1 point

Sample Answer 1

$XSLDF = LDF^{unlim} \times (1 - R^L) / (1 - R_{12}^L)$

Ultimate $R^L = 1.21 / 2.27 = 0.533$

$R_{12}^L = y = 0.533 \times e^{0.177 \times 3} = 0.906$

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$$\text{XSLDF} = 3.707 \times (1 - 0.533) / (1 - 0.906) = 18.417$$

$$\text{Ult XS Losses} = 18.417 \times (6,984 - 6,216) \times 1,000 = 14,144,256$$

Sample Answer 2

$$\text{Ult Ratio} = 1.21 / 2.27 = 0.533 = \text{ILF}^{250K} / \text{ILF}^{1M}$$

$$Y \text{ for AY2014} = 0.533 \times e^{0.177(3)} = 0.906 = R_t^L$$

$$\text{LDF} = R_t^L (\text{LDF}^L) + (1 - R_t^L) \text{XSLDF}^L$$

$$3.707 = (0.906) (2.261) + (1 - 0.906) \text{XSLDF}^L$$

$$\text{XSLDF}^L = 17.644$$

$$\text{Ult} = (6,984,000 - 6,216,000) (17.644) = \$13,550,592$$

Sample Answer 3

$$\text{XSLDF}_t^L = \text{LDF}_t \times (1 - R_t^L) / (1 - R_t^L)$$

$$\text{Ult ratio} = R_t^L = (1 - 0.47) / (1 - 0.5) = 0.558$$

$$\text{At } t=12 \text{ months, } y = \text{Ult ratio } e^{0.177 \times 3} = 1.70 \times \text{Ult ratio} = 1.70 \times 0.558 = 0.948$$

$$\text{XSLDF}_t^L = 3.707 \times (1 - 0.558) / (1 - 0.948) = 18.417$$

$$\text{Current losses in layer} = 6,984,000 - 6,216,000 = 768,000$$

$$\text{Ult losses in layer} = 768,000 \times 31.77 = 24,399,360$$

Sample Answer 4

$$U = 12,860 / 24,106 = 0.533$$

$$\text{XSATU} = \text{ATU} \times (1 - R_{48}^L) / (1 - R_{12}^L) = 3.708 \times (1 - U e^0) / (1 - U e^{0.177 \times 3}) = 18.508$$

$$\text{XS Ult} = (6,984K - 6,216K) \times 18.508 = 14,213K$$

Sample Answer 5

$$\text{Ult ratio} = (6,984K \times 3.7079) / (6,216K \times 2.2616) = 0.5429$$

$$\text{Ratio}^{250/1000 @ 12 \text{ months}} = \text{Ult ratio} \times e^{0.177t} = 0.5429 \times e^{0.177 \times 3} = 0.92337$$

$$\text{LDF} = 3.7079 \times (1 - R_{ULT}) / (1 - R_{12}) = 3.7079 \times (1 - 0.5429) / (1 - 0.9233) = 22.0975$$

$$(6,984,000 - 6,216,000) \times 22.0975 = 16,970,880$$

Sample Answer 6

$$\text{Ult ratio} \times e^{0.177 \times 3} = 6,216 / 6,984$$

$$\text{Ult ratio} = 0.523$$

$$\text{LEV}_{250-1000} / \text{LEV}_{1000} \text{ at 12 months} = 768 / 6,984$$

$$\text{LEV}_{250-1000} / \text{LEV}_{1000} \text{ at ult} = 1 - 0.523$$

$$\text{CDF} = 3.698 \times (1 - 0.523) / (768 / 6,984) = 16$$

$$16 \times (6,984 - 6,216) = 12,288$$

EXAMINERS' REPORT

Part a

Candidates performed very well on this part in general, with a majority receiving full credit. This part related to knowledge statements about methods for estimating unpaid claims in a layer excess of a retention but bounded by a limit, and how to apply per-occurrence charges in particular. The three key steps were to determine expected losses at \$250k limit, to determine expected losses at \$1M limit, and to take the difference to obtain expected losses in the \$750k excess of \$250k layer.

A few candidates attempted to use ILFs instead of per-occurrence charges to calculate losses by layer, which doesn't work as the highest ILF for losses above \$1M is not provided in the problem. Other candidates included an extraneous $(1.0 - 0.47)$ multiplied against the 0.05 per-occurrence charge at

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\$1M, even though the per-occurrence charges apply strictly to ground-up losses. Finally, a few candidates attempted to include an aggregate loss charge in the calculations, even though no aggregate loss coverage was indicated in the problem.

Part b

Candidates performed very well on this part in general, with a majority receiving full credit. This part related to knowledge statements about methods for estimating unpaid claims in a layer excess of a retention but bounded by a limit, and how to estimate and apply loss development factors for losses in different loss layers. The key steps were to determine cumulative LDFs at a \$250k loss limit, determine cumulative LDFs at a \$1M loss limit, multiply those cumulative LDFs by reported losses for AY 2014 at a 12 month evaluation, and take the difference between those calculated ultimate layer losses to project the ultimate losses in the \$750k excess of \$250k loss layer.

A few candidates mistakenly used ILFs in the formulas instead of calculating LDFs from the loss development triangles provided, or else they mistakenly applied both LDFs and ILFs. A few other candidates subtracted 1.0 from each of the cumulative LDFs when applying the LDFs to the reported losses at either or both of the \$250k and \$1M limits, which corresponded to the candidate providing IBNR for the excess layer rather than the requested ultimate losses. In other cases, there were mathematical errors made in the computation and application of the interval LDFs, but given that LDFs needed to be computed for six different interval/limit combinations, those mathematical errors were fairly infrequent.

Part c

This tested the relationship of development patterns between layers.

In order to obtain full credit, candidates needed to figure out how losses limited at 250K and 1000K relate to each other, both at 12 months of development and at ultimate.

As a second step, they needed to find the loss development factor from 12 months to ultimate for the losses within that layer and finally, apply that LDF to the reported losses at 12 months.

This question was challenging and a significant number of candidates were unable to provide a meaningful response.

Another portion of candidates were not able to provide a good calculation for the first two steps, but they provided a calculation of ultimate losses in the layer using the correct amount or reported losses in the layer (given in the question) with whatever LDF they came up with, earning partial credit.

Other candidates were able to fully answer the question, but made an error in deriving the ratio of reported losses limited at 250K to reported losses limited at 1M at various development periods. The variable t in the formula is defined as being the time to ultimate and, while the proper exponent in the formula for t at 12 months development is $4 - 1 = 3$, several candidates used $t=1$.

Candidates used several different approaches to solve the problem:

- Figuring out the ratio of losses at ultimate can be calculated using Excess charges, ILF, actual reported losses from AY 2011 or from ultimate losses calculated in part (b).
- Figuring out the ratio of losses at 12 months can be calculated using the formula given in the question but also directly from the reported losses at 12 months.
- Finally, there are 2 possible equations that can be used to calculate the LDF to ultimate for the losses in the layer.

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All these possible variances lead to several combinations of acceptable methods, with final answers varying within a range from \$12 million to \$24 million.